



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-PH101/BSPH101 Physics-I

UPID : 001003

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 × 10 = 10]

- (I) Write an expression for Compton Wavelength of an electron.
- (II) MB statistics is applicable for photons. State True or False.
- (III) What is the displacement of a particle executing SHM, when the velocity is maximum?
- (IV) What is population inversion?
- (V) For which type of materials magnetic susceptibility is negative?
- (VI) Phase velocity and group velocity are equal when the medium is _____
- (VII) Which statistical distribution is followed by free electrons in a conductor?
- (VIII) Write the differential equation for SHM.
- (IX) What is Optic Axis of a crystal?
- (X) What is the dimension of $\mu_0 \epsilon_0$?
- (XI) An electron and a proton are accelerated through the same potential difference. The de Broglie wavelength of the proton is larger than that of the electron. State True or False.
- (XII) Give one example of Uniaxial Crystal.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 × 3 = 15]

2. a) Draw the plot for blackbody radiation against wavelength for three different temperatures $T_1 < T_2 < T_3$ [5]
b) What do you mean by Ultra-Violet Catastrophe?
3. (a) Show that the gradient of a scalar field $u(x,y,z)$ is perpendicular to the surface $u(x,y,z)=\text{constant}$ [5]
(b) Find Gradient of the following scalar field at the point (1, -2, -1).
 $u(x,y,z) = xy^2z + 4x$
4. State Brewster's Law. Calculate the polarizing angle for light traveling from water to glass. Refractive index of glass=1.53 and water =1.33 [5]
5. Differentiate between Amplitude resonance and velocity resonance. What is logarithmic decrement in damped oscillation? [5]
6. A particle is trapped in an infinite one dimensional potential well of width 'L'. Write its wavefunction and probability density function in its first excited state. Find the probability density at $x=L/2$. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 × 3 = 45]

7. (a) Show that the gradient of a scalar function is an irrotational vector, [3]
(b) Show that Curl of position vector is zero. [3]
(c) Is the magnetic field an irrotational vector or a solenoidal vector? Is the electric field an irrotational vector or a solenoidal vector? [2]
(d) Solve the following . [4]
If $\psi(x,y,z) = 3xy^2 - 5x^2z + 2z^2$, find $\nabla^2 \psi$
(e) Determine the constant 'q' so that the following field is solenoidal [3]
 $\vec{F} = (2x + 3yz)\hat{i} + (3y - z^2)\hat{j} + (y^3 + qz)\hat{k}$
8. (a) Distinguish between Fresnel and Fraunhofer Diffraction [3]
(b) Write the characteristics of LASER. [4]

- (c) Mention three methods of producing plane polarized beam of light. [3]
- (d) Explain briefly what do you mean by double refraction. [3]
- (e) State Malus law. [2]
9. (a) What is Grating element? Write and plot the intensity distribution of a single slit diffraction pattern and find the positions of maxima, minima and secondary maxima. [2+1+2+4]
- (b) Find the minimum thickness of a half wave plate and a quarter wave plate for a light wave of wavelength 5893 Angstrom. Given $\mu_0 = 1.66$ and $\mu_e = 1.47$ [2]
- (c) What are the three basic components of a LASER? What is an optical resonator? [2+2]
10. (a) X-Ray photon of wavelength 2 angstroms is scattered at an angle of 45 degrees by a carbon block. [2+3]
- (i) Calculate the wavelength of scattered X ray
- (ii) Find the fraction of energy lost by the X-Ray photon during this event.
- (b) A particle of mass m is confined in an infinite potential well between $x=0$ and $x=L$. Solve the time independent Schrodinger equation for such a system and Find the normalized eigenfunctions and eigenvalues. Using the normalized eigenfunctions find the expectation value of x . [5+2]
- (c) Plot the Fermi distribution of a particle at $T=0$ K and $T>0$ K. [3]
11. (a) Calculate the number of photons emitted per sec by a 20 W source of monochromatic light of wavelength 431 nm. [4]
- (b) What do you mean by degeneracy? <https://www.makaut.com> [2]
- (c) State the Heisenberg's Uncertainty Principle. The average time interval that elapses between the excitation of an atom and the time it radiates energy is 10^{-8} seconds. Calculate the uncertainty in energy of emitted photons. [2+3]
- (d) Compare Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac Statistics on the basis of minimum three characteristics. [4]

*** END OF PAPER ***